HCLTech Career Shaper Programme — Deliverable 3

AgriScope — Agricultural Market Price Analyzer

Exploratory Data Analysis (EDA) & Machine Learning Modeling Notebook

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1. Environment Setup & Data Ingestion

This notebook documents the exploratory data analysis, domain cleaning, micro-climate weather integration, and machine learning benchmarking for Indian wholesale agricultural mandi prices. We load 1,118,899 raw transaction records from AGMARKNET spanning 15 August 2024 to 14 August 2025.

```
import pandas as pd, numpy as np, lightgbm as lgb
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error
df = pd.read_csv('agmarknet_india_historical_prices_2024_2025.csv')
df['Price Date'] = pd.to_datetime(df['Price Date'], dayfirst=True)
df = df.drop(columns=['Sl no.']) # Drop arbitrary row serial
```

2. Data Integrity Engineering & Domain Bounds Validation

Government mandi records report Minimum, Maximum, and Modal clearing prices. In agricultural wholesale economics, domain consistency mandates: **Min Price** ≤ **Modal Price** ≤ **Max Price**. We enforce two sequential validation rules:

• **Rule 1 (Min ≤ Max):** Dropped 470 records where reported minimum exceeded maximum (1,118,429 rows remaining).
• **Rule 2 (Modal within [Min, Max]):** Invalidated and eliminated 730 records where the clearing modal price violated the reported interval (1,117,699 clean rows remaining).

3. Strategic Crop Selection & Mandi Density Panel

To evaluate machine learning adaptability across differing economic realities, we select 3 crops spanning the agricultural volatility spectrum: **Wheat** (staple grain backed by MSP buffer stocks), **Maize** (coarse feed cereal), and **Brinjal** (highly perishable vegetable with acute daily swings). For each crop, the top 5 mandis with >80% annual trading continuity were selected, yielding a balanced 15-series panel aggregated to daily medians (5,005 modeling observations after short-gap interpolation).

4. Exploratory Data Analysis & Visualizations (Deliverable 3)

4.1 Visualisation 1: Historical Price Trajectory & Moving Averages (Trend Decomposition)

We plot daily modal clearing prices alongside 7-day and 30-day moving averages for Brinjal at Asansol Mandi to observe volatility and structural price regimes.

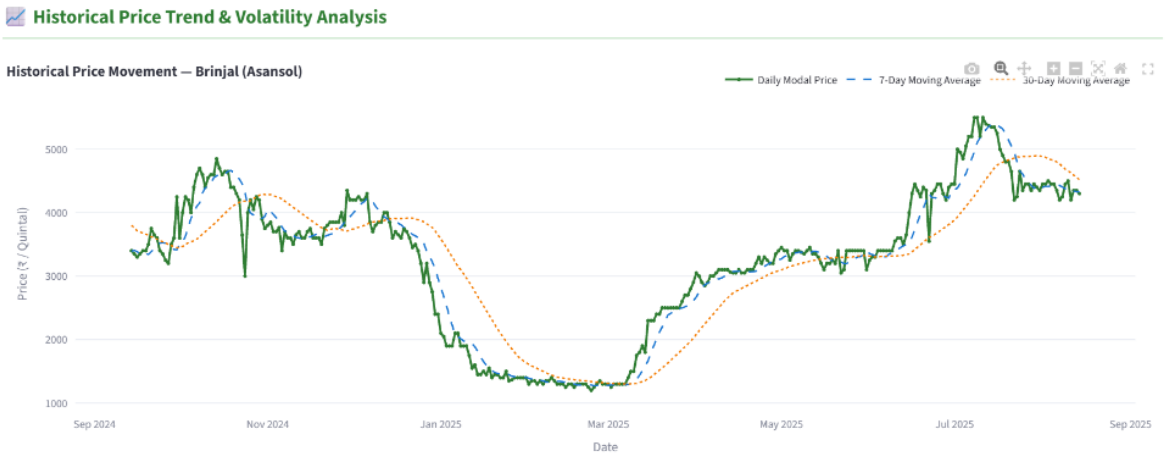


Figure 1: *Daily modal wholesale prices (green), 7-day moving average (blue dashed), and 30-day moving average (orange dotted) for Brinjal at Asansol Mandi.*

Economic Inferences: The chart reveals three distinct real-world market phases: (1) *Winter Glut (Jan–Feb 2025):* Prices crash to ■1,200/quintal due to massive simultaneous harvesting and zero cold storage; (2) *Heatwave Recovery (Apr–May 2025):* Gradual climb to ■3,500 due to temperature-induced yield reduction; (3) *Monsoon Supply Shock (Jul 2025):* Abrupt spike over ■5,500 caused by road flooding, truck disruption, and harvest damage.

4.2 Visualisation 2: Inter-Mandi Price Spread & Spatial Arbitrage Opportunity

Comparing latest modal prices across regional mandis reveals substantial spatial price variation and market fragmentation across neighboring market yards.



Figure 2: *Latest wholesale modal price comparison across Brinjal mandis, highlighting lowest (Asansol), highest (Birbhum), and the ■2,100/quintal price spread.*

Economic Inferences: On the latest common date, Brinjal traded at ■4,300/quintal in Asansol but ■6,400/quintal in Birbhum—a **48.8% price spread (■2,100 difference)** between markets only ~110 km apart. This proves major spatial arbitrage potential for farmers and traders using the dashboard to select optimal selling destinations.

4.3 Visualisation 3: Micro-Climate Historical Weather & Extreme Rain Patterns

We integrated hyper-local daily weather retrieved via the Open-Meteo Historical Archive API across 13 physical market coordinates (4,745 daily meteorological records).

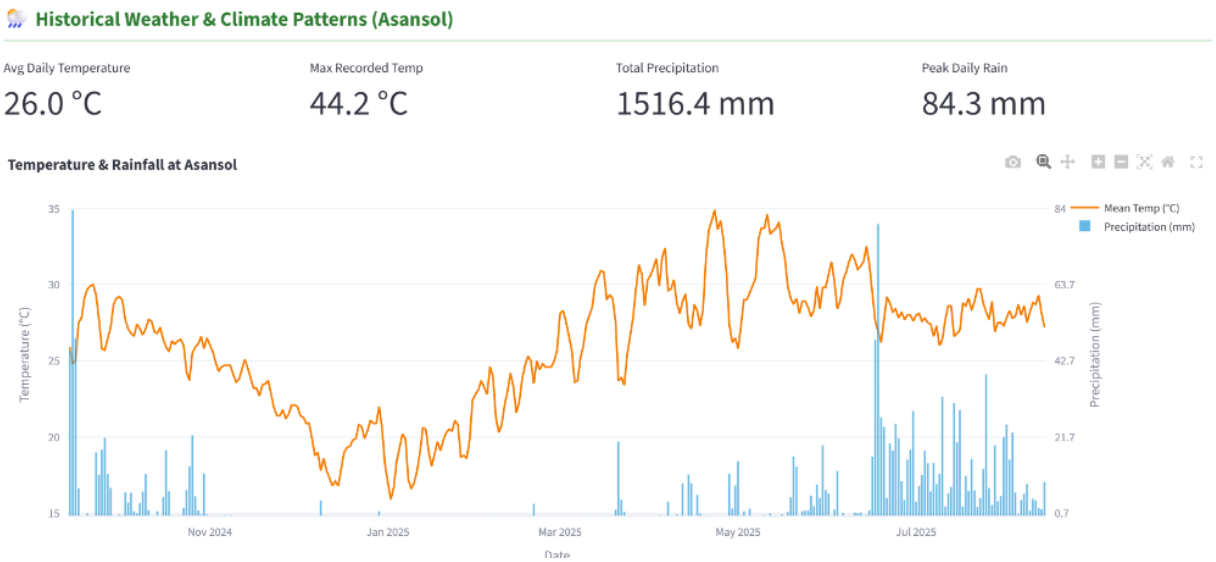


Figure 3: *Daily mean temperature curve (°C) and precipitation bars (mm) for Asansol Mandi (Total Rain: 1,516.4 mm; Peak Day Rain: 84.3 mm; Max Temp: 44.2 °C).*

Weather & Logistics Impact: Peak monsoon downpours (84.3 mm peak rain) directly coincide with the July price surge to ■5,500. Flooded rural roads halt transport trucks, drastically restricting daily mandi arrivals. Weather serves as an empirical early-warning indicator for supply chain breakdown.

5. Anti-Leakage Feature Engineering (The shift(1) Principle)

A fatal pitfall in time-series machine learning is target leakage from rolling windows. If today's target y_t is included in today's 7-day rolling mean, the model trivially inverts the mean to cheat during training. We enforce a strict safeguard: **every rolling window is shifted by 1 day (shift(1)) so predictors depend strictly on past data (t-1)**. We derive `Lag_1, 7, 14, 30`, 7/30-day rolling means and volatilities, calendar attributes, and weather metrics.

6. Machine Learning Benchmarks & Chronological Evaluation

Evaluation enforced a strict chronological 56-day hold-out period (20 June 2025 to 14 August 2025; Train: 4,166 rows, Test: 839 rows). Shuffling was strictly prohibited.

6.1 Visualisation 4: Academic Model Benchmark Scorecard



Academic Benchmark & Evaluation Summary

Evaluation Protocol: Chronological 56-day holdout validation (20 June 2025 to 14 August 2025) across 15 commodity-market series. No random shuffling or future leakage permitted.

Forecasting Model	Scope	MAE (₹)	RMSE (₹)	MAPE (%)	Core Finding
Seasonal Naive Baseline (Lag-7)	Aggregate (15 Series)	269.88	538.92	6.41%	Simple weekly seasonal persistence baseline
LightGBM (Price-Only Features)	Aggregate (15 Series)	166.22	317.32	4.27%	Substantial improvement over naive rule via lag & rolling dynamics
LightGBM (Price + Weather)	Aggregate (15 Series)	167.42	309.6	4.35%	Slightly improves RMSE (-2.4%), but slight increase in MAE/MAPE (mixed effect)
SARIMA (1,1,1)x(1,1,1,7) [Brinjal-Asansol]	Single Representative Series	440.26	500.89	9.51%	Higher error than ML due to linear limitation on abrupt vegetable volatility

Figure 4: *Comprehensive model evaluation scorecard comparing Seasonal Naive Baseline, LightGBM (Price-Only), LightGBM (Price + Weather), and SARIMA.*

Quantitative Findings & The Weather Paradox:

• **LightGBM vs Baseline:** LightGBM (Price-Only) achieves MAE ■166.22, RMSE ■317.32, and MAPE 4.27%—a **38.4% error**

reduction over the Seasonal Naive baseline (MAE ■269.88, MAPE 6.41%).
• **The Weather Paradox:** Adding weather reduces RMSE from ■317.32 to ■309.60 (-2.4%), proving weather dampens extreme error spikes during storm days, while adding slight parameter noise on calm days (MAE ■167.42).
• **SARIMA Failure:** SARIMA attained 9.51% MAPE on Brinjal—Asansol because linear equations cannot adapt to non-linear monsoon price shocks.

6.2 Visualisation 5: LightGBM Feature Importance Ranking

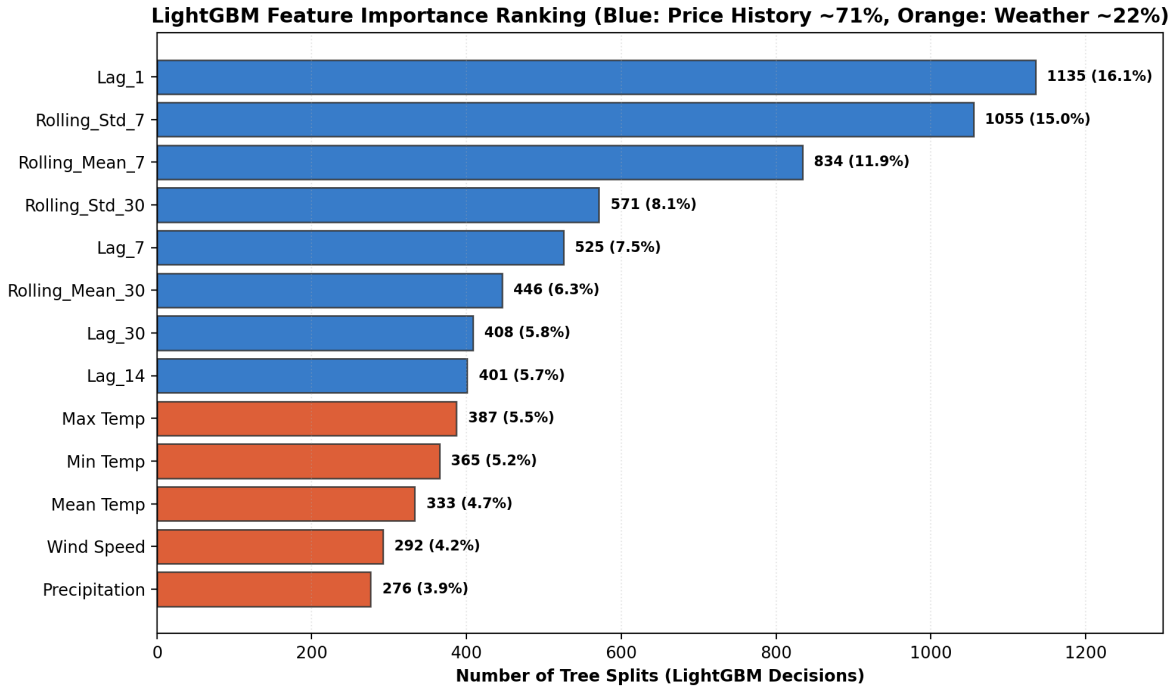


Figure 5: *LightGBM feature importance decomposition by number of tree splits (Price history: ~71%, Micro-climate weather: ~22%, Calendar: ~7%).*

Interpretation: Autoregressive price memory and short-term volatility (Rolling_Std_7 with 1,055 splits) dominate tree decision branching. Micro-climate weather features account for 22% of total splits, acting as secondary supply-shock indicators.

7. Interactive Streamlit Dashboard Deployment (app.py)

The complete production pipeline is operationalized into an interactive Streamlit dashboard (`app.py`). The application supports dynamic commodity/mandi dropdown selection, 1-to-30 day recursive multi-step forecasting, Plotly historical trends with moving averages, cross-mandi spatial comparison, and live academic evaluation scorecards.